

Nov 5

What function does the power series  $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{2^{n+2}}$  represent?

↳ Remember that  $\frac{a}{1-r} = \sum_{n=0}^{\infty} ar^n$  ( $|r| < 1$ )

↳ within the radius of convergence

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{2^{n+2}}$$

Transform to Geom since we know what to do w/ that

$$= \sum_{n=0}^{\infty} \left( \frac{-1}{2} \right)^n \left( \frac{x}{2} \right)^n$$

$$a = \frac{-1}{2}$$

$$r = \frac{x}{2}$$

↳ will converge as long as

$$|x| < 2$$

Since GS converges for  $|r| < 1$

When  $|r| < 1$ , converges at  $\frac{a}{1-r} \rightarrow \frac{\frac{-1}{2}}{1 - \frac{x}{2}} = \frac{-1}{2(1 - \frac{x}{2})} = \frac{-1}{2 - x}$

So the equivalent function is  $\frac{1}{2-x}$  when  $|x| < 2$